

BATHURST, AustraliaWMO: **947290**Lat: **33.4119S**Lon: **149.6539E**Elev: **745**StdP: **92.69**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.7	-1.5	-6.2	2.4	13.5	-4.4	2.9	8.6	10.8	9.2	9.7	9.3	1.2	180	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	33.7	18.2	31.5	17.5	29.7	17.1	20.6	27.7	19.8	26.3	19.1	25.3	5.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.0	21.6	18.0	14.2	21.2	17.2	13.4	21.0	62.8	27.4	59.8	26.4	57.3	25.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	8.7	7.7	-5.0	36.5	1.1	2.4	-5.8	38.2	-6.5	39.6	-7.1	40.9	-7.9	42.6
			-5.2	22.1	1.1	0.9	-6.0	22.7	-6.7	23.2	-7.3	23.7	-8.1	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	21.3	20.2	17.6	13.5	9.5	7.1	6.2	7.3	10.3	13.2	16.5	19.1
	DBStd	5.95	3.32	3.17	2.84	2.90	2.66	2.33	2.08	2.32	2.83	3.27	3.70	3.36
	HDD10.0	386	0	0	0	6	42	90	119	90	30	8	1	0
	HDD18.3	2056	11	14	48	146	274	336	376	342	242	160	77	30
	CDD10.0	1647	349	286	236	112	26	4	1	6	38	108	197	283
	CDD18.3	276	102	67	24	2	0	0	0	0	2	23	55	
	CDH23.3	3379	1160	673	287	42	0	0	0	0	7	84	385	741
	CDH26.7	1283	523	257	72	5	0	0	0	0	0	10	144	271
(14) Wind	WSAvg	3.3	3.6	3.5	3.2	2.8	2.8	3.0	3.2	3.4	3.5	3.4	3.6	3.6
(15) Precipitation	PrecAvg	654	75	63	50	43	47	35	47	56	49	64	60	64
	PrecMax	989	224	236	157	166	115	142	125	163	125	216	196	173
	PrecMin	214	1	1	1	0	7	1	4	2	8	3	7	0
	PrecStd	164	55	57	39	39	28	25	29	36	26	40	37	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	35.9	31.4	27.2	21.0	16.5	16.0	19.7	24.3	28.0	33.9	34.6
		MCWB	19.0	19.1	17.1	15.4	12.9	11.0	9.9	11.2	13.0	14.6	17.2	17.4
	2%	DB	34.2	32.2	28.7	24.3	19.0	14.7	14.0	17.0	21.9	25.7	30.6	31.7
		MCWB	18.5	18.3	17.0	14.3	12.1	10.4	8.8	9.9	12.2	14.0	15.7	17.2
	5%	DB	31.8	29.8	26.5	22.3	17.2	13.4	12.6	15.2	19.6	23.6	27.7	29.7
		MCWB	18.1	17.6	16.3	13.6	11.5	9.7	8.6	9.1	11.6	13.6	15.5	16.6
	10%	DB	29.7	27.4	24.4	20.4	15.6	12.2	11.2	13.3	17.4	21.3	24.9	27.6
		MCWB	17.6	17.5	16.0	13.1	11.0	9.3	7.9	8.5	10.8	13.1	15.2	16.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.9	20.9	17.5	15.6	13.0	12.5	13.0	15.3	17.5	19.4	20.3
		MCDB	29.5	29.7	24.9	22.0	17.4	14.3	13.7	16.4	20.1	22.5	26.3	26.9
	2%	WB	20.6	20.5	19.1	16.1	13.9	11.6	10.5	11.3	13.9	16.2	18.2	19.4
		MCDB	28.6	27.5	23.4	20.7	16.4	13.1	12.1	14.5	18.5	21.7	25.3	25.9
	5%	WB	19.9	19.6	18.2	15.1	12.7	10.6	9.2	10.1	12.8	15.1	17.2	18.5
		MCDB	27.0	25.7	22.9	19.9	15.9	12.5	11.3	13.8	17.2	20.4	24.0	25.0
	10%	WB	19.1	18.8	17.2	14.1	11.6	9.6	8.3	8.9	11.7	14.1	16.2	17.6
		MCDB	26.1	24.5	22.2	18.8	14.9	11.8	10.6	12.7	16.0	19.5	22.3	24.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	12.8	12.8	13.5	12.2	9.9	10.2	11.7	12.9	13.5	13.9	14.3
		MCDBR	18.0	16.8	16.4	16.4	14.9	12.2	13.1	15.5	16.9	17.5	18.8	17.6
		MCWBR	5.8	5.8	6.7	7.4	8.6	8.0	8.5	8.9	8.7	8.0	7.0	6.1
		MCDBR	13.9	12.8	12.4	13.2	11.4	9.9	10.3	12.7	13.4	13.8	14.2	13.9
	5% WB	MCWBR	5.5	5.4	6.0	6.7	7.2	7.0	7.3	8.0	7.8	7.5	6.6	6.0
(40) Clear-Sky Solar Irradiance	taub		0.345	0.342	0.319	0.297	0.283	0.273	0.267	0.275	0.304	0.319	0.341	0.333
	taud		2.502	2.538	2.576	2.609	2.593	2.639	2.646	2.614	2.513	2.496	2.475	2.525
	Ebn at Noon		992	975	964	934	895	884	909	946	963	987	990	1008
	Edh at Noon		114	106	95	82	73	66	68	80	99	109	116	112
(44) All-Sky Solar Radiation	RadAvg		7.14	6.15	5.18	4.13	3.12	2.38	2.73	3.64	5.01	6.09	6.60	7.27
	RadStd		0.46	0.55	0.48	0.36	0.21	0.20	0.20	0.26	0.38	0.57	0.63	0.51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.62	N/A	N/A	+1.77	N/A	N/A	N/A	-129	+206	+106
Regional (36 neighbors)	+0.44	N/A	N/A	+0.84	N/A	N/A	-2	-88	+189	+99

Nomenclature: See separate page